

Page Replacement Algorithms

- **Page Fault** happens when a program tries to use a page that is **not present in RAM** and is stored in **swap space (disk)**.
- When this happens, the **Operating System (OS)** must **bring the required page from disk into RAM**.
- If there is a **free frame** in RAM, the OS **loads the page directly** into that frame.
- If the system is **highly utilized** and **all frames are full**, the OS must **remove one existing page** to create space.
- The **page replacement algorithm** decides **which page should be removed** from RAM.
- The selected old page is **swapped out to disk**, and the **new page is swapped into the freed frame**.

In short:

Page not in RAM → Page Fault → OS loads page → RAM full → Old page replaced using page replacement algorithm

Types of Page Replacement Algorithms

(Main AIM: Minimum Page Faults)

1. FIFO (First In First Out)

Meaning:

- The **page that comes first into memory is removed first**, just like a queue.

How it works:

- Pages are loaded into frames in order.
- When memory is full, the **oldest page is replaced**, even if it is still important.

Advantages:

- **Very easy to understand and implement.**

Disadvantages (Why performance is not good):

- The page removed might be:
 - **A startup/initialization page** that is no longer needed → *good replacement*
 - OR a **very important page** that was loaded early and is **used again and again** → *bad replacement* → causes **more page faults**

Real-world example:

- Imagine a **waiting line at a temple**.
- The person who entered first is forced to leave first, **even if he is still praying**.

- That makes no sense, but FIFO works exactly like this.

Belady's Anomaly (Only in FIFO)

- Normally, **more memory frames = fewer page faults**.
- But FIFO sometimes behaves strangely:
 - **Increasing frames can increase page faults.**
- This strange behavior is called **Belady's Anomaly**.
- **LRU and Optimal do NOT show this problem.**

2. Optimal Page Replacement

Meaning:

- Replace the page that **will not be used for the longest time in the future**.

How it works:

- If a page will **never be used again**, replace it.
- If all pages will be used again, replace the page that is **needed farthest in future**.

Advantages:

- **Gives the minimum possible page faults.**
- It is the **best algorithm in theory**.

Disadvantages:

- **Impossible to implement practically**, because the OS **cannot know future page requests**.

Real-world example:

- While packing a bag for a trip, you remove the item
 - that you **know you won't need again**.
- Perfect decision, but only possible if you **know the future**.
- Similar to **SJF scheduling** in CPU.

3. LRU (Least Recently Used)

Meaning:

- Replace the page that **has not been used for the longest time in the past**.

Logic:

- **Recent past $\approx$ near future**
- If a page hasn't been used for a long time, it is **less likely to be used again**.

Implementation Methods:

(a) Counter Method

- Each page has a **time value**.
- Whenever a page is used, its time is updated.
- The page with the **smallest time value** (oldest use) is replaced.

(b) Stack Method

- Maintain a **stack of page numbers**.
- When a page is used:
 - Remove it from its position
 - Put it on **top of the stack**
- **Top → Most Recently Used**
- **Bottom → Least Recently Used**
- **Doubly linked list** is used because removal happens in middle.

Real-world example:

- Books on your study table:
 - Books you used **recently** stay on top
 - Books you haven't touched for months go to the bottom
- When space is needed, you remove the **least recently used book**.

4. Counting-Based Page Replacement

- OS keeps a **counter** for each page showing **how many times it is used**.

(a) LFU – Least Frequently Used

Meaning:

- Replace the page with the **smallest reference count**.

Logic:

- Pages used many times are **important**.

Real-world example:

- In a shop, remove items that **customers rarely buy**.

(b) MFU – Most Frequently Used

Meaning:

- Replace the page with the **largest reference count**.

Logic:

- A page with a **small count may be new** and not yet used enough.

Real-world example:

- You remove a **popular product** assuming customers already bought it enough.

Why LFU and MFU are not common?

- Reference count alone is **not reliable**.
- Old pages may have high count but are **no longer useful**.
- Hence, **LRU is preferred in practice**.